

Resume — Inhyeok Hwang

INHYEOK HWANG

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SUMMARY

Mechanical engineering researcher with an M.S. from Konkuk University's Reliability-Based Design Optimization (RBDO) Lab, 4 peer-reviewed publications (IEEE TIM · Solar Energy · Journal of Power Electronics, 2024; PHM Asia-Pacific, 2023), and 1 patent (co-inventor). My research spans IPMSM motor control modeling and fault physics simulation — FEM-based fault implementation (Ansys Maxwell), inverter Co-simulation (MATLAB/Simulink), and Sim2Real validation — through to accelerated life testing, PHM/RUL estimation, and lifetime prediction under varying stress environments, across three distinct material systems (semiconductor, polymer, motor drivetrain).

At GINT Corp., I have applied this academic foundation across three years of industrial product development: from motor control firmware through full APQP lifecycle management to first-article production plan. The bridge between academic rigor and production reality is where I operate.

Beyond domain depth, I bring a demonstrable record of autonomous learning and technical writing. My three publications span three structurally distinct failure mechanisms — semiconductor bond-wire degradation, polymer photovoltaic lifetime, and electromechanical drivetrain fault diagnosis — each acquired from primary literature within a single 2.5-year M.S. research period. That breadth of structured domain acquisition is exactly the kind of cross-domain capability that AI training data curation demands.

Core identity: ME researcher with publication depth · autonomous learner · technical writer · industrial practitioner.

PUBLICATIONS & PATENTS

PEER-REVIEWED JOURNALS (3, CO-AUTHORED)

Year	Journal	Title (abbreviated)	Role
2024	IEEE Transactions on Instrumentation and Measurement , Vol. 73	Bond-wire fault detection in IGBT via inverter output parameters	3rd author
2024	Solar Energy (Elsevier), Vol. 276, Art. 112645	Lifetime prediction of PV polymers under varying environments via damage summation (Miner's Rule)	4th author
2024	Journal of Power Electronics , Vol. 24(5), pp. 822–831	Failure mode identification in IPMSM under ALT via dual sensor architecture	6th author

CONFERENCE PRESENTATIONS (4)

Year	Conference	Topic	Award
2023	PHM Society Asia-Pacific	120 kW IPMSM system-level fault diagnosis	—
2022	PCIM Asia	IGBT IPM quasi-DC power cycling	—
2022	Korean Society for Reliability (KSR)	Predictive maintenance for EV drivetrain	Best Paper Award
2021	Korean PHM Society	IGBT open-fault diagnosis via system-level measurements	Best Poster Award

PATENTS

- **Patent #1** — Motor initial position detection · Co-inventor · KR App. PN231067KR (2023) · Applicant: GINT Inc. · Detection time reduced: 1.0 s → **0.56 s**
- **Patent #2** — *Method for preventing step-out of rotor of motor in low temperature and apparatus thereof* · KR App. **10-2023-0175484** (filed 2023-12-06) · Reg. **10-2654562** (granted 2024-04-01) · Applicant: GINT Inc. · IPC H02P 29/032 · Development contributor (experimental validation; not on inventor list) · −40°C cold-start time: 300 s → **100 s**

EDUCATION

KONKUK UNIVERSITY — M.S. IN MECHANICAL DESIGN (RBDO LAB)

Mar 2021 – Feb 2023 · 27 major credits · Major GPA 3.77/4.5 · Advisor: Prof. Namsu Kim

Thesis: *Fault Diagnosis Simulation of Interior Permanent Magnet Synchronous Motor using Electromagnetic Analysis* (49 pp., RISS: f678963f23f2e418ffe0bdc3ef48d419)

- Constructed an IPMSM digital twin: implemented fault modes (static/dynamic eccentricity, winding asymmetry) in Ansys Maxwell FEM; coupled with MATLAB/Simulink inverter control model for Co-simulation; reproduced fault signatures (phase current harmonics, flux density distribution) under multiple operating conditions and validated against experimental data (Sim2Real precursor methodology)
- Participated in government-funded PHM SoC development project (MOTIE, Apr 2021 – Oct 2022): RUL estimation for IPMSM drivetrain fault diagnosis using multi-sensor signal fusion
- Observed full ALT design-test-redesign cycle for IPMSM eccentricity fault characterization — load factor application, FTA, dual-sensor architecture operation (P-05, J. Power Electron. 2024, 6th author)
- Contributed to Power Cycling Test work on IGBT modules characterizing bond-wire lift-off as a primary failure mode — online detection via inverter output parameters (V_{ce} saturation, switching energy) without in-circuit instrumentation (P-01, IEEE TIM 2024, 3rd author, DOI 10.1109/TIM.2024.3472910)
- Research experience: 2.5 years total (6 months undergrad research assistant + 2 years M.S.)

KONKUK UNIVERSITY — B.S. IN MECHANICAL DESIGN

Mar 2015 – Feb 2021 · 79 major credits · Major GPA 3.24/4.5

EXPERIENCE

GINT CORP. — ELECTRICAL CONTROL R&D + PRODUCT DEVELOPMENT (JUNIOR PM, CONCURRENT)

Senior Researcher · Jan 2025 – Present

GT-SS500: 48V Electric Autonomous Speed Sprayer — Full Lifecycle from R&D to Mass Production

- Managed full product lifecycle from algorithm research through first-article production plan of 16 units plus 2 transport carts and 2 spares (APQP Phase 2–3), coordinating across 6 cross-functional teams (design / manufacturing / sourcing / quality / safety / sales)
 - Authored 5 Design FMEAs covering Step 1–7; identified 5 high-priority action items (AP=H), all resolved before production ramp
 - Led 4 root-cause analyses to closure — MCB electrolytic corrosion, control-mode state machine fault, pump freeze O-ring failure, GND bounce noise
 - Designed and built 4 custom test rigs in-house (dynamometer, fan bench, pump bench, bumper safety rig)
 - Operated 27-item NCR closed-loop corrective action process; defined IQC/OQC criteria for supplier qualification
 - Designed 5-node CAN distributed control architecture (VCU ↔ dual drive MC ↔ fan ESC ↔ pump DRV ↔ BMS); authored 4 DBC files; standardized 3-pass consistency validation protocol
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GINT CORP. — ELECTRICAL CONTROL R&D

Researcher · Feb 2023 – Dec 2024

EOP 400W: Automotive BLDC Motor Control (Government R&D Project)

- Developed SVPWM/DPWM switching strategy — reduced FET temperature 1–6°C, input power 1–3.8% (validated on 2,932 data points)
 - Demonstrated cold-start improvement at –40°C: 300 s → 100 s (subject of Patent #2; experimental validation conducted independently)
 - Built dynamometer torque-control system achieving 0.008% nonlinearity; presented at internal engineering seminar
 - 1 patent as co-inventor + development contribution to 1 additional patent (see Patents section above)
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SKILLS

MECHANICAL ENGINEERING DOMAINS

- Motor design & control: IPMSM, BLDC, sensorless FOC, SVPWM/DPWM
- System architecture: CAN distributed control (5-node network design, DBC authoring, consistency validation)

- Reliability engineering: ALT (participated in full cycle as observer), PCT (directly conducted), DFMEA (Step 1–7), root-cause analysis, PHM/RUL estimation
- Simulation & modeling: FEM (Ansys Maxwell), Co-simulation (MATLAB/Simulink), digital twin, Sim2Real validation
- Test infrastructure: custom test rigs designed and built in-house — torque-control dynamometer (0.008% nonlinearity), fan airflow bench (flow velocity measurement), pump endurance rig, bumper safety rig (stop distance / peak current / settling time)
- Lifetime analysis: Damage Summation (Miner's Rule), Weibull estimation (concept), multi-environment stress mapping

RESEARCH & COMMUNICATION

- Peer-reviewed journal publications: IEEE TIM (Vol. 73, 3rd author), Solar Energy (Vol. 276, 4th author), J. Power Electronics (Vol. 24, 6th author)
- Conference presentations with awards: KSR Best Paper Award (2022), Korean PHM Society Best Poster Award (2021)
- Technical writing: journal manuscripts, patent specifications, design documentation, 1,900+ lines of portfolio materials
- Autonomous learning: self-directed cross-domain domain acquisition across three distinct failure mechanisms (semiconductor · polymer · electromechanical) within a single M.S. research period; self-directed acquisition of APQP/DFMEA methodology from primary standards documentation during industrial transition
- Bilingual: Korean (native) · English (professional reading & writing)

TOOLS & PLATFORMS

- Programming: Python, MATLAB/Simulink, C (embedded — STM32, NXP MC9S12ZVMC)
 - Signal processing & ML familiarity: multi-sensor PHM, prognostics models, classical anomaly detection
 - Documentation: Markdown, LaTeX, Git workflow
 - Instrumentation: CANoe, Vector VN1600, NI DAQ (NI-9215), oscilloscope
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SELECTED RESEARCH HIGHLIGHTS

- **IGBT bond-wire fault detection via inverter output parameters** — Power Cycling Test work characterizing failure mechanism at system level without in-circuit instrumentation (IEEE TIM 2024, 3rd author)

- **PV polymer lifetime prediction under continuously varying environments** — Damage Summation (Miner's Rule) applied to multi-environment stress histories; demonstrates ability to structure complex multi-variable ME problems rigorously (Solar Energy 2024, 4th author)
- **Failure mode identification in IPMSM under ALT via dual-sensor architecture** — Observed full design-test-redesign cycle; methodology illustrates ALT principles applicable to AI training explanation (J. Power Electronics 2024, 6th author)
- **120 kW IPMSM system-level fault diagnosis** — Multi-sensor signal fusion for drivetrain RUL estimation (PHM Society Asia-Pacific 2023)
- **IPMSM digital twin thesis** — Implemented eccentricity and winding fault modes in Ansys Maxwell FEM; co-simulated with MATLAB/Simulink inverter control model to reproduce fault signatures under both normal and faulted operating conditions; validated simulation outputs against experimental measurements (Sim2Real methodology) (Konkuk University, 2023)